Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 8

Time Allowed: 30 minutes

General Instructions:

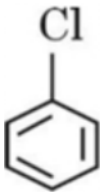
Read the following instructions carefully.

1. There are 14 questions in this question paper with internal choice.
2. SECTION A consists of 46 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

BHAYAT

Section A

1. Which of the following is most reactive towards nucleophilic substitution reaction? **[1]**

a) 

b) CH_3-CH_2-CH_2-Cl

c) CH_2=CH-CH_2-Cl

d) CH_2 = CH-C

2. The deficiency of which of the following vitamins causes Scurvy? **[1]**

a) Vitamin B_{12}

b) Vitamin C

c) Vitamin A

d) Vitamin B_6

3. Which of the following reagents can be used to oxidise primary alcohols to aldehydes? **[1]**

a) CrO_3 in an anhydrous medium.

b) Pyridinium chlorochromate.

c) All of these

d) Heat in the presence of Cu at 573K.

4. Which of the following compounds will give a ketone on oxidation with chromic anhydride (CrO_3)? **[1]**

a) $CH_3CH_2CH_2OH$

b) $(CH_3)_3C$-OH

c) $CH_3-CH_2-\underset{\underset{OH}{|}}{C}H-CH_3$

d) $(CH_3)_2CH$-CH_2OH

5. The half life of a substance in a first order reaction is 15 min. The rate constant is **[1]**

a) 4.62×10^{-2} min^{-1} b) 2.46×10^{2} min^{-1}

c) 7.18×10^{2} min^{-1} d) 6.74×10^{-2} min^{-1}

6. Match the items given in column I with that in column II.

Column I	Column II
(a) Molarity	(i) $\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$
(b) Molality	(ii) Number of gram moles of a solute per litre of solution
(c) Normality	(iii) Number of gram moles of a solute per kg of solvent
(d) ppm	(iv) Number of gram equivalent of a solute per litre of solution

a) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i) b) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

c) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv) d) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

7. C - Cl bond in chlorobenzene in comparison to C - Cl bond in methyl chloride is: **[1]**

a) longer and weaker b) longer and stronger

c) shorter and stronger d) shorter and weaker

8. Which of the following elements of 3d series of transition elements has the lowest $\Delta_a H^0$? **[1]**

a) Zn b) Cu

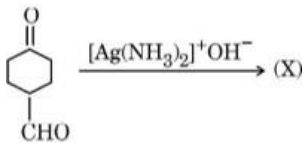
c) Sc d) Cr

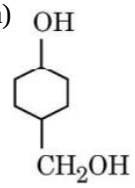
9. As temperature increases, the reaction rate: **[1]**

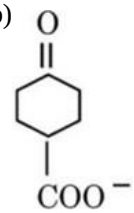
a) Stays the same b) Increases

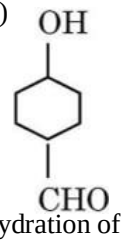
c) First decreases then increases d) Decreases

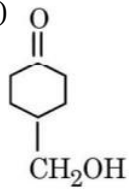
10. The product (X) obtained in the following reaction is: **[1]**



a) 

b) 

c) 

d) 

11. Dehydration of tertiary alcohols with Cu at 573 K gives: **[1]**

a) Ketone b) Alkyne

BHAYAT

lpropanamide into 1- phenylethanamine is .
[1]

[1]

a) Br_2/NaOH

b) $LiAlH_4$

c) excess H_2/Pt

d) $NaBH_4$/methanol

13. **Assertion (A):** The backbone of DNA and RNA molecules is a chain consisting of heterocyclic base, pentose sugar and phosphate group.

Reason (R): Nucleotides and nucleosides mainly differ from each other in presence of phosphate group.

a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

c) Assertion (A) is true, but Reason (R) is false.

d) Assertion (A) is false, but Reason (R) is true.

14. **Assertion (A):** Benzoic acid does not undergo Friedel - Crafts reaction. **[1]**

Reason (R): Carboxyl group is deactivating and the catalyst aluminium chloride gets bonded to the carboxyl group.

BHAYAT

a) Both A and R are true and R is the correct explanation of A.

b) A is false but R is true.

c) Both A and R are true but R is not the correct explanation of A.

d)A is true but R is false

BHAYAT